NUS GRIP
Ambient Systems

Assignment #3

September 29, 2020

Assignments 1-2 Revisited

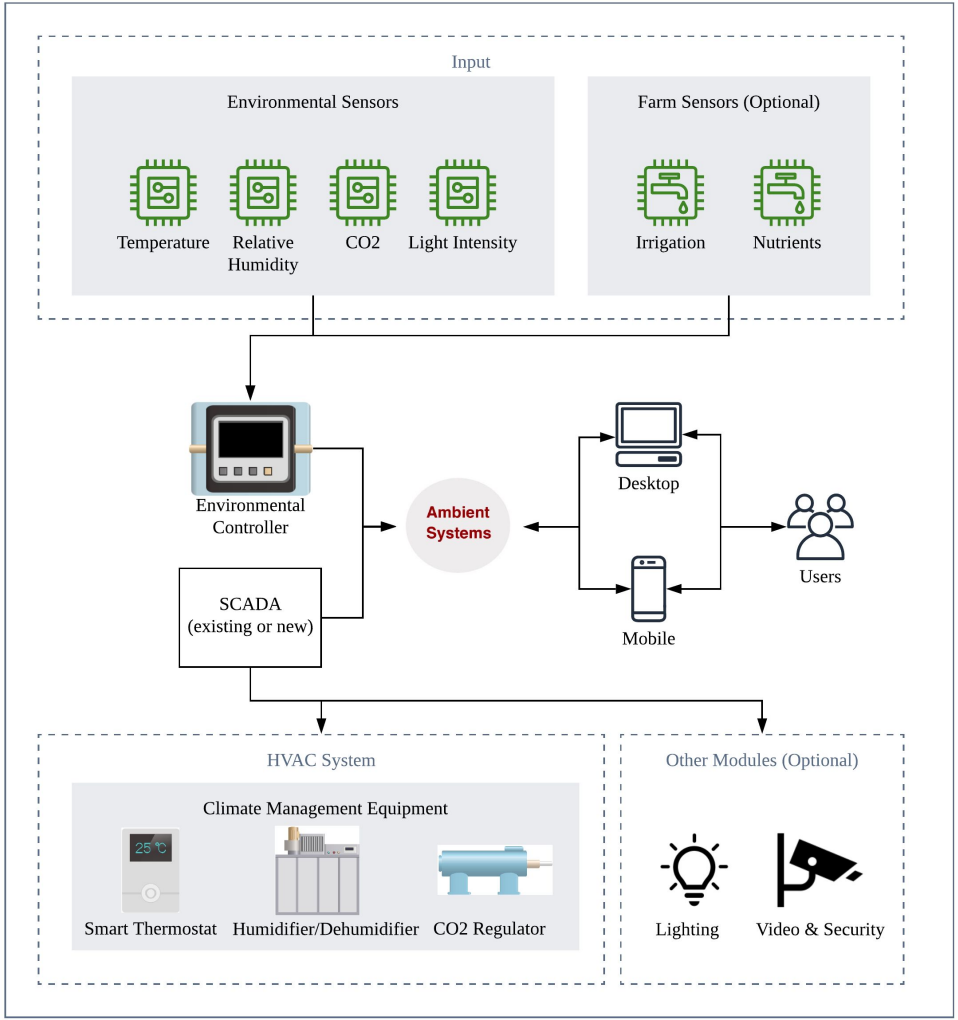


MVP: 3-part solution



IoT Sensor Network	Software Platform	Service Model
• Temperature sensor (0 to 50 C +/- 0.1 C) • Relative humidity sensor (0% to 100% +/- 1%) • CO2 sensor (ppm +/- 5 ppm) • Light intensity sensor • Relevant connectors, mountings, gateway, cabling, and controls	• Real-time data delivered using modern user interface design • Provides alerts via email or text when environmental conditions are beyond optimal range • Provides recommendations to remedy the situation	• Provide a “Climate-as-a-Service” • User pays an annual subscription fee based on farm type and area • Streamline financing of the CAPEX and OPEX of the system

Ambient MVP High-level Schematic



MVP: Two Paths

Path	Original Equipment Manufacturer (OEM)	System Integrator
Activities	1. Design IoT device based on 2. Procure components 3. Assemble device with casing 4. Develop firmware 5. Install on-site	1. Procure controller and sensors 2. Install on-site 3. Connect with existing systems
Advantages	• Flexible customization • Financially sustainable	• Ready to use • Manufacturer support
Disadvantages	• Electronics design cost and complexity • Manufacturing and distribution costs	• Solutions are wired • No data analytics capabilities • Manufacturers are far away


Device Enclosure


Temp sensor

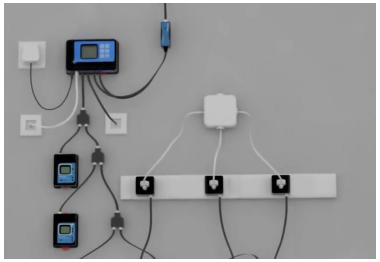

CO2 sensor


Micro-controller (“the platform”)


RH sensor



Transceiver


Connected controller and sensors

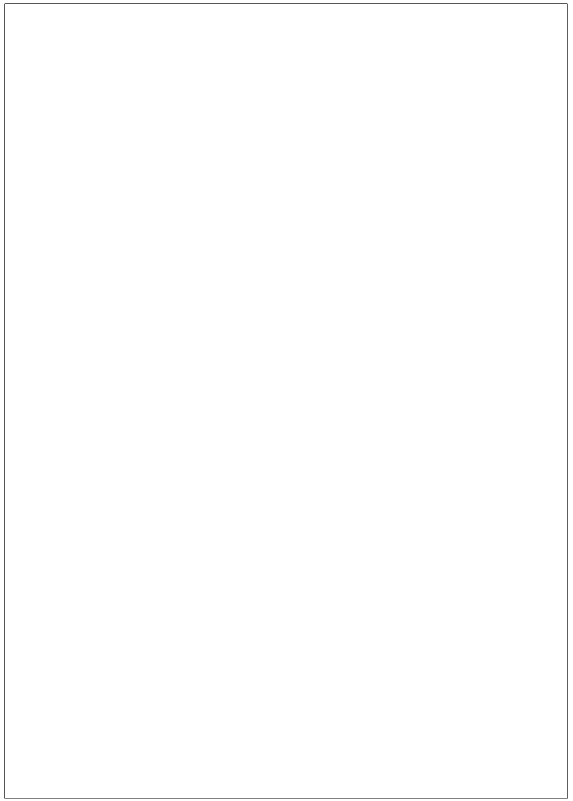

Controller


Sensors

MVP: Model

- System lays down the foundation for the prediction and optimization model
- Numerous options (depends on requirements)
- Potential: neural network (supervised learning)
- Consists of several “layers” of data -- potential structure:
  - Layer one: sensor measurements (i.e., raw sensor data)
  - Layer two: equipment operational logs
  - Layer three: output (yield, harvest time, crop quality)
- Use-cases:
  - Receive recommendations for climate controls (e.g., adjust temperature, increase humidity)
  - Predict crop output, harvest time, and quality
  - Automate climate controls

IP and



is an Intelligent Building Automation System, Method, and Apparatus for Smart Building Applications, which:

- Functions as an intelligent building management and automation system
- Manages indoor climate using Internet of Things (IoT) and
- Uses a diverse real-time dataset collected from an efficient and secure IoT network of sensors
- Trains -enabled models from IoT data to predict climate conditions and optimize the environment
- Provides a dashboard with real-time data to increase visibility into the climate conditions, along with user alerts when climate is not within target range
- Connects with existing Supervisory Control and Data Acquisition (SCADA) system used for building management
- Provides instructions to Heating, ventilation, and air conditioning (HVAC) equipment connected to SCADA
- Automates the building's HVAC system and related equipment to bring the indoor air quality within the desired goal

Section 1: Quality System and Certification



Standards and Certifications

Singapore	Europe and US	International
IMDA: ● TR 47 : 2016 IoT reference architecture for Smart Nation ● TR 50 : 2016 IoT information and services interoperability for Smart Nation ● IMDA IoT Cyber Security Guide (2019) ● MTCS Certification Scheme	CISPR 22 standard for Information Technology Equipment-Radio Disturbance Characteristics-Limits and Methods of Measurement. More specifically: ● EN55022 ● EN61000-4-2 ● EN61000-4-3 ● EN55024 CE marking Federal Communications Commission (FCC) ● Part 15 Class A	ISO/IEC: ● 90003:2014: Software engineering ● 27000:2018: Information technology — Security techniques — Information security management systems

Section 2: BoM and Expenses



Bill of Materials: IoT Sensors

Component Name	Part Number	Manufacturer	Description	Picture	Unit Price	Supplier	Quantity	Total
Board and microcontroller	Arduino Uno Rev3 SMD -- A000066	Arduino	Board with processor and attachment ports		$29.86	RS-Online	2	$59.72
Temperature Sensor	MAX31889ALT+	Maxim Integrated	Board Mount Temperature Sensors 0.1 degrees C accuracy, 1-Wire Temperature Sensor		$4.40	Mouser	2	$8.80
Relative Humidity Sensor	SHT31-DIS-F2.5kS	Sensirion	Board Mount Humidity Sensors RH/ T Sensor		$9.11	Mouser	2	$18.22
CO2 Sensor	T6713-6H	Amphenol Advanced Sensors	Air Quality Sensors CO2 Sensor Module for AAS-AQS-UNO		$122.54	Mouser	2	$245.08
Light Sensor	TSL2561T ams	ams	TSL2561T ams, Ambient Light Sensor Unit Surface Mount 6-Pin TMB		$4.41	RS-Online	2	$8.82
Transceiver	LR1110IMLTRT	Semtech	RF Transceiver LONG RANGE WI-FI LOC LORA TRANSCEIVER		$14.93	Mouser	2	$29.86
ADCs	AD7795BRUZ	Analog Devices	Analog Devices AD7795BRUZ, 16-Bit Serial ADC Differential Input, 24-Pin TSSOP		$16.02	RS-Online	8	$128.16
Device enclosure	NSYTBP342912T	Schneider Electric	Plastic Enclosure, Flame Retardant, Multipurpose, Polycarbonate, 341 mm, 291 mm, 128 mm, IP66		$106.51	Element 14	2	$213.02
							Total	**$720 for 2 sensors**

Detailed bill of materials for the OEM MVP path has been been made for IoT Sensors

Bill of Materials: Variable (Device + Install)

Component	Quantity	Unit	Unit Cost	Total Cost
Variable				
Sensor: Combined Temp, RH, CO2, and Light	2	#	$400.00	$800.00
Mountings	2	#	$50.00	$100.00
Data Cables (e.g., CAT6) and Accessories	25	m	$10.00	$250.00
Power Cables and Accessories	25	m	$5.00	$125.00
Gateways / Routers / Switches (Internet connection)	2	#	$100.00	$200.00
Other hardware: CCTV system	4	#	$100.00	$400.00
Technician Contractors (Electrical/Mechanical)	24	hrs	$200.00	$4,800.00
Cloud computing software subscriptions	1	monthly usage bill	$500.00	$500.00

$7,175.00

Expenses (2021 Monthly)

Fixed Costs (Monthly in Year 1)

Working Space and Utilities	1	Monthly Rental Fee	$2,000.00	$2,000.00
Front-end Design and Development (UX/UI for web and mobile)	100	hrs	$100.00	$10,000.00
Application Server Development (Database and model)	100	hrs	$100.00	$10,000.00
IoT Device Development (Device Assembly and Firmware Programming)	100	hrs	$100.00	$10,000.00
Marketing Agency	50	hrs	$100.00	$5,000.00
Working Equipment (e.g., MVP studio equipment rental)	1	Monthly Rental Fee	$1,000.00	$1,000.00
IP Development and Licensing	1	Monthly Fee	$1,000.00	$1,000.00
Team: Project Manager	1	Monthly salary	$6,000.00	$6,000.00
Team: Senior Engineer (Electrical/Electronics/Mechanical)	1	Monthly salary	$6,000.00	$6,000.00
Team: Senior Software Engineer	1	Monthly salary	$6,000.00	$6,000.00
Team: Business Development Associate	1	Monthly salary	$3,000.00	$3,000.00
			Monthly	**$60,000.00**
			Annual	$720,000.00

Section 3: Pricing



Pricing: TEV and PV

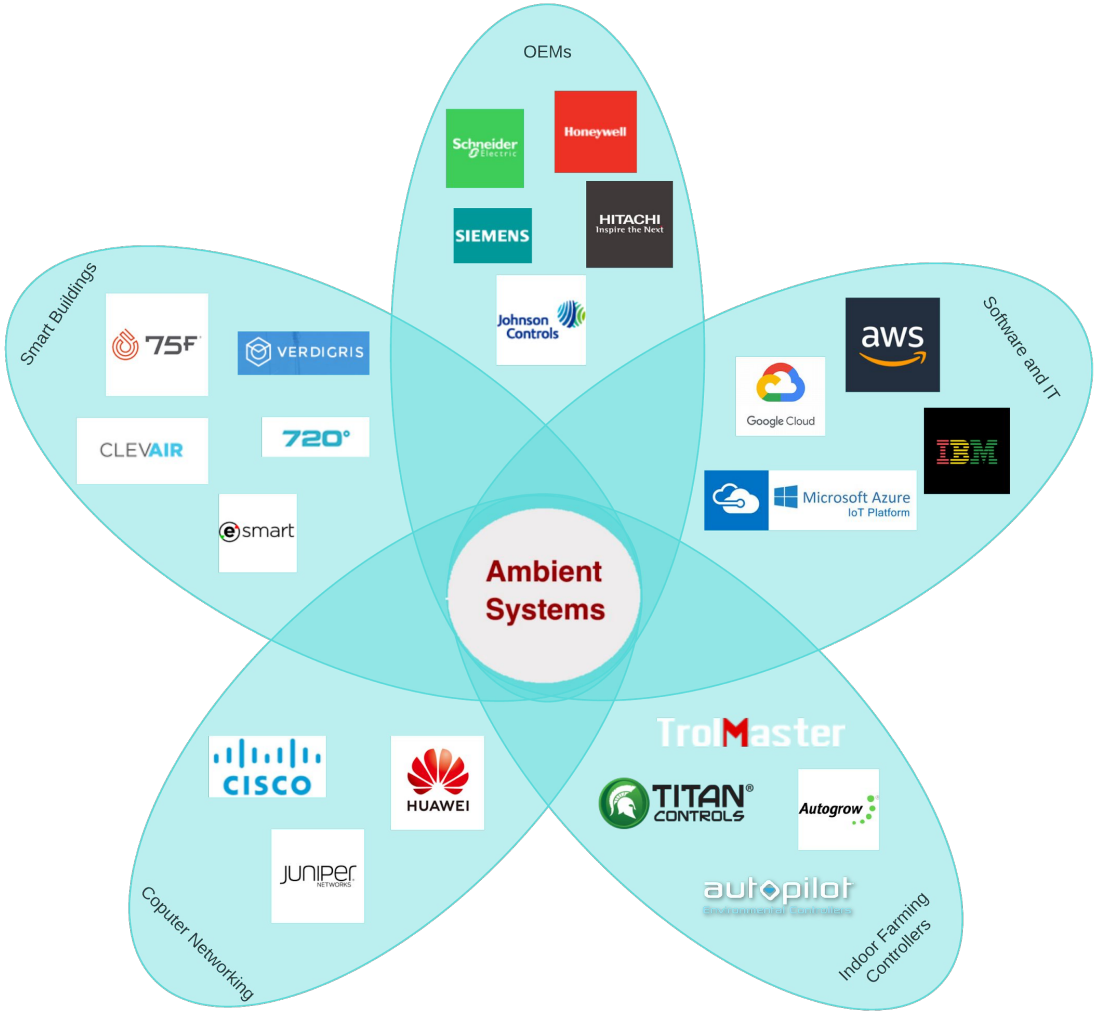
	Best Alternative	New Product
Hardware Cost	$3,000.00	$0.00
Installation Time (hrs)	40	0.00
Connection Time (hrs)	40	0.00
Lifetime (months)	60	60
Data Management Time (hrs) (10 hrs per month for 5 years)	600	0.00
System Management Time (hrs) (100 hrs per year for 5 years)	500	0.00
Hourly Personnel Cost	$75.00	$0.00
Real Price	**$96,000.00**	**$14,871.80**
Cost per m2	**$3,227.59**	**$500.00**
Cost per month	**$1,600.00**	**$247.86**

Section 4: Competitive Analysis



Competitive Landscape

There are several companies in the smart buildings space, but there is still a need for a tailored solution for vertical farms.



Section 5: Business Model Canvas



Value Proposition Canvas: Vertical Farms

Business Model Canvas



Key Partners

1. NUS
2. Farm Owners and Operators
3. Building Owners and Operators
4. HVAC & BMS Vendors
5. Government Agencies



Key Activities

Requirements Gathering
Platform Development
System Installation



Key Resources

IP
Platform
ML Model



Value Proposition

Initial: Enable successful climate management of vertical farms by measuring key parameters and ensuring process repeatability for future harvests.

With data: increase yield, optimize energy use, and reduce labour costs.



Customer Relationships

Tailored Service



Channels

Direct Outreach



Customer Segments

1. Farm Owners and Operators
2. Building Owners and Operators



Cost Structure

CAPEX: Solution Design, and Software Development

OPEX: Software Maintenance and Improvement, Project Management (System Installation)



Revenue Streams

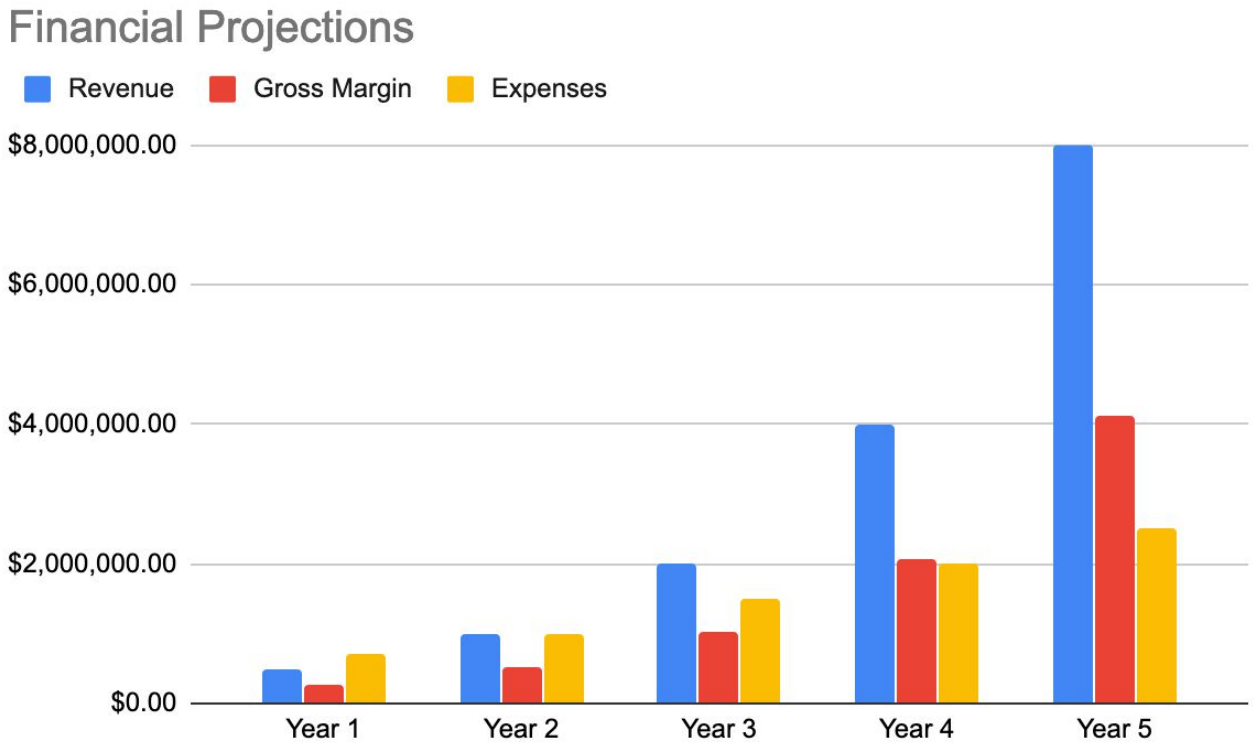
Subscription Fee (Climate-as-a-Service)

Section 6: Financial Projections



Preliminary Financial Projections

~$1.4M Funding Gap



Preliminary Financial Projections

~$1.4M Funding Gap

Inputs

Product price ($/m^2)	500
Product COGS ($/m^2)	242

	Year 1	Year 2	Year 3	Year 4	Year 5
Revenue	$500,000.00	$1,000,000.00	$2,000,000.00	$4,000,000.00	$8,000,000.00
Area under management (m2)	1000	2000	4000	8000	16000
minus Cost of Goods Sold	$242,000.00	$484,000.00	$968,000.00	$1,936,000.00	$3,872,000.00
Gross Margin	$258,000.00	$516,000.00	$1,032,000.00	$2,064,000.00	$4,128,000.00
Expenses	$720,000.00	$1,000,000.00	$1,500,000.00	$2,000,000.00	$2,500,000.00
Profit & Loss = Net Cashflow	-$462,000.00	-$484,000.00	-$468,000.00	$64,000.00	$1,628,000.00

Cumulative Net Cashflow	-$204,000.00	$32,000.00	$564,000.00	$2,128,000.00	$5,756,000.00

Thank you